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Adaptive Cruise Control for Cut-In Scenarios Based on Model Predictive Control Algorithm

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Abstract: In a cut-in scenario, traditional adaptive cruise control usually cannot effectively identify the cut-in vehicle and respond to it in advance. This paper proposes an adaptive cruise control (ACC) strategy based on the MPC algorithm for cut-in scenarios. A finite state machine (FSM) is designed to manage vehicle control in different cut-in scenarios. For a cut-in scenario, a method to identify and quantify the possibility of cut-in of a vehicle is proposed. At the same time, a safety distance model of the cut-in vehicle is established as the basis for the state transition of the finite state machine. Taking the quantified cut-in possibility of a vehicle as a reference, the model predictive control (MPC) algorithm, which considers the constraints of driving safety and comfort, is used to realize coordinated control of the host vehicle and the cut-in vehicle. Simulink–Carsim simulation studies show that the ACC strategy for a cut-in scenario can effectively identify a cut-in vehicle and quantify the possibility of cut-in of the vehicle. Faced with a cut-in vehicle, the host vehicle using the ACC strategy can brake several seconds early and switch the following target to the cut-in vehicle. Meanwhile, the acceleration and jerk of the host vehicle changes within a reasonable range, which ensures driving safety and comfort.

Keywords: adaptive cruise control; cut-in possibility; safe distance model; model predictive control

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1. Introduction

With the rapid development of computers, electronics, and sensor technology, advanced driver assistance systems (ADAS) have now become an important direction for development of the automotive industry because they have great advantages in improving transportation efficiency [1] and driving safety [2]. As an important branch of ADAS, ACC systems have a wide range of applications in car-following scenarios. However, complex driving scenarios, such as cut-in scenarios, are still challenging for ACC systems [3]. When facing the lane-change maneuvers of a nearby car, a vehicle with an ACC system has to switch the following target to the cut-in vehicle and follow its trajectory. In this process, there may be a problem of braking too late as well as excessive braking. In order to ensure driving safety and driving comfort, the cut-in possibility of a nearby car needs to be determined. When facing a vehicle cut-in, the host vehicle with the ACC system needs to cooperate with the cut-in vehicle at the same time, and smoothly follow the cut-in vehicle.

In ACC control, the aim is to solve the optimal control law of the host vehicle under the condition of satisfying constraints. Therefore, ACC can be considered as a constrained optimization problem. For constrained optimization methods, Garone et al. [4] and Nicotra et al. [5] proposed a novel scheme called the explicit reference governor (ERG) for the control of constrained systems characterized by fast dynamics and that are subject to non-convex constraints. Hosseinzadeh et al. [6] proposed a systematic approach for applying the ERG to linear systems subject to combinations of intersections and unions of concave constraints. In addition, Forsgren et al. [7] first utilized the barrier functions in constrained optimization. In constrained optimization, a barrier function is a continuous

function in which the value of a point increases to infinity as the point approaches the boundary of the feasible region; therefore, a barrier function is used as a penalizing term for violations of constraints [8]. However, in the process of following a preceding vehicle, the motion state of the preceding vehicle changes dynamically, so the control method of the host vehicle needs to be continuously adjusted. Model predictive control (MPC) can divide the time domain into infinite finite horizons. In each horizon, which is a sample time, the system experiences an error with the reference trajectory. With this error minimization as the control objective, the optimal control law is solved under constraints, and then the control law is used as the input of the system at the next horizon until the system reaches the reference trajectory. This is considered a rolling optimization process. Therefore, MPC is more suitable for the following control of a host vehicle. In addition, MPC can take the acceleration of the preceding vehicle as a disturbance in the system control, which compensates for the uncertainty caused by the preceding vehicle. For instance, Bageshwar et al. [9] proposed an MPC-based headway control algorithm, which explicitly incorporates acceleration limits to meet the requirements of ride comfort and safety.

Except for MPC, many studies have contributed to the ACC control method. Milanés et al. [10] proposed the design, development, implementation, and testing of a cooperative adaptive cruise control system. Yi et al. [11] analyzed an ACC system using the method of transition state variables and obtained a desired acceleration. Mobus et al. [12] studied the motion characteristics of an adaptive cruise system using a non-linear method with finite time constraints. Shakouri et al. [13] optimized the non-linear characteristics of a vehicle into linear characteristics, thereby using the linear time-invariant model to control the ACC tracking index. Li et al. [14] conducted real-time research and observations on a vehicle dynamic model, and optimized the adaptive cruise system algorithm by using the minimum tracking method. Paul et al. [15] designed a full-speed-range ACC system using the fuzzy control theory. The genetic algorithm, particle swarm algorithm, and differential evolution algorithm have been used to optimize the parameters of the fuzzy controller. The designed ACC system can be used in urban traffic and highways. Considering the time-consuming problem of adaptive control design, caused by the combination of different characteristics and situation-dependent behaviors, Naus et al. [16] proposed a systematic method of parameterized adaptive control design, based on explicit model predictive control. Schakel et al. [17] investigated the driving state of a driver when the driver did not use the ACC function for comparison experiments, and collected information, such as acceleration and vehicle spacing, which was taken in the form of observations of lane changes.

In addition to the study of ACC control strategies, many studies have taken the driving characteristics of a driver in the design of ACC into account. Wang et al. [18] developed an adaptive longitudinal driving assistance system based on driver characteristics using the recursive least squares method to identify the parameters of the driver model based on manual operation, and applied those parameters to real-time control. Lefèvre et al. [19] used a set of offline driving data to teach a driver model instead of a human driver, and proposed a new intelligent longitudinal speed control (LVC) framework, which included a driver model and a model predictive controller. Kuderer et al. [20] applied a feature-based demonstration learning method, in which the driver model is trained to produce a driving trajectory, similar to that shown by a driver, during vehicle following. Lefevre et al. [21] designed a self-learning driver model that used the hidden Markov model (HMM) and Gaussian mixture model (GMM) to identify and predict a driver's steering intention and acceleration, combining it with the MPC algorithm to design an ACC system controller. The system could adapt to different types of drivers and ensure driving safety. Butakov et al. [22,23] also designed different assisted driving strategies for ADASs, based on different types of drivers. Bifulco et al. [24] proposed a full-speed ACC; by activating the electronic control unit (ECU), they used a few minutes of the online learning mode to learn a driver's preferences and attitudes, and then updated the parameters to achieve a full-speed ACC that conformed to the driver's behavior characteristics. Although the above

papers show various studies on ACCs, the cut-in scenarios are still a challenge. Therefore, an ACC control strategy needs to consider cut-in possibility, and controls a vehicle to make a decision accordingly, which is meaningful to improve ACC driving safety and comfort.

The ACC for cut-in scenarios based on the model predictive control algorithm in this paper is structurally illustrated in Figure 1. To begin this process, a finite state machine (FSM) was adopted to introduce different cut-in scenarios, which generated a reference trajectory for vehicle control. The second method used to recognize potential cut-in vehicles and to quantify the cut-in possibility based on nearby vehicle states was proposed. By comparing, judging the position of a potential cut-in vehicle with the position of the host vehicle, a potential cut-in vehicle on both sides of a lane was determined. Through analysis of the lateral displacement and lateral velocity of a potential cut-in vehicle, the cut-in possibility of a potential cut-in vehicle was quantified and then used as the reference for MPC to realize a coordinated control between the host vehicle and the cut-in vehicle. In addition, considering a straight lane and a curved lane, a safety distance model of the cut- in vehicle was proposed in this paper. When a nearby vehicle cuts into the lane, by considering the state change among the cut-in vehicle, the host vehicle and the preceding vehicle in front of it in the lane, a safety distance model of the cut-in vehicle was established, which is used to manage the transition of the FSM. In the end, considering the constraints of driving safety, comfort, and taking the possibility of cut-in of a nearby vehicle as a reference, an ACC control strategy based on MPC (MPC–ACC) was developed. Through the control strategy proposed in this paper, the host vehicle can judge and recognize a cut-in vehicle, and, at the same time, for a cut-in vehicle in different states, the host vehicle is controlled in coordination with the MPC–ACC to achieve safe and smooth processes in cut-in scenarios. The contributions of this paper are the following aspects:

(1) A finite state machine is used to divide the cut-in scenarios, and a method of screening and quantifying potential cut-in vehicles is proposed, which can be used as a reference for vehicle control to cooperate with a cut-in vehicle.

(2) The safety distance model of the cut-in vehicle is proposed. Comprehensively considering the movement state of the vehicle, the preceding vehicle ahead of the host in the lane as well as the cut-in vehicle, the safety distance model of the cut-in vehicle is established, which performs the conversion of the management finite state machine so that vehicle control can be taken appropriately when a cut-in vehicle changes lanes.

(3) Facing the cut-in vehicles in different states, the MPC–ACC control strategy is developed to realize the follow-up control for cut-in vehicles. At the same time, driving safety and comfort are taken as the constraints of the MPC–ACC to make the host vehicle follow an optimized control trajectory.

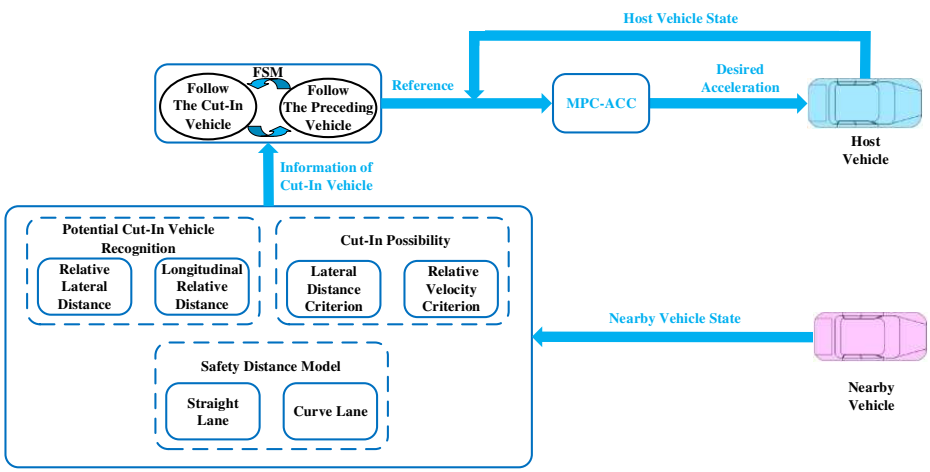

Figure 1. MPC–ACC control strategy architecture for cut-in scenarios.

The rest of this paper is as organized follows. Cut-in vehicle identification and quantification are shown in Section 2. Section 3 describes the cut-in vehicle safety distance model. Section 4 introduces the MPC–ACC control strategy model. In Section 5, the simulation result research is given. The conclusions are presented in Section 6.

2. Cut-In Scenario Classification and Cut-In Possibility Quantification

Cut-in scenarios are divided by the FSM according to the state of the surrounding vehicles. With the presence of nearby vehicles on both sides of a host vehicle, whether the vehicles cut in is judged and the cut-in possibility is quantified, so as to control the host vehicle accordingly.

2.1. Finite State Machine for Cut-In Scenarios

The finite state machine is shown in Figure 2. The host vehicle can adopt different control strategies when facing a cut-in vehicle through the conversion of the FSM. The FSM includes two states in total, in which the cut-in vehicle state contains two sub-states; corresponding to three scenarios.

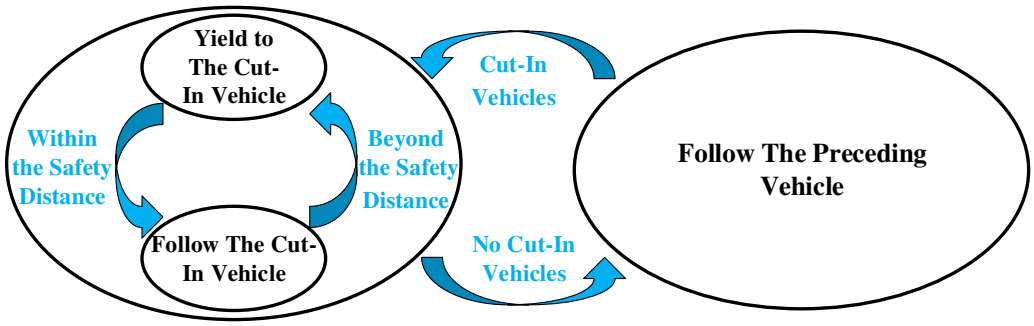

Figure 2. The FSM of the host vehicle.

Driving scenario 1: When the cut-in vehicle does not exist, or completes a lane change, the FSM remains in the following the preceding vehicle state, which means that the host vehicle can use the preceding vehicle in front of it in the lane as a reference to stably follow the object.

Driving scenario 2: When the host vehicle recognizes a potential cut-in vehicle using the target screening method mentioned below, if the longitudinal relative distance (which can be measured using a corner radar equipped on the front of the host vehicle) between the host vehicle and the preceding vehicle in front of it in the lane meets the cut-in vehicle safety distance model, the FSM is switched to follow the cut-in state. At this time, the host vehicle does not need to brake in advance to maintain a safe relative distance with the cut-in vehicle. When the cut-in vehicle reaches the center line of the lane, the host vehicle stably changes the follow object, from the preceding vehicle, to the cut-in vehicle.

Driving scenario 3: When a nearby vehicle performs a cut-in maneuver, and the relative distance between the host vehicle and the cut-in vehicle does not meet the safety distance model, the FSM is switched to yield to the cut-in vehicle state. Before the nearby vehicle cuts in, the host vehicle changes the following target to the cut-in vehicle. At the same time, the cut-in possibility is used as a reference trajectory for the MPC. At this time, the host vehicle under the control of the MPC–ACC start to perform a braking action, which increases the relative distance from the preceding vehicle in front of it in the lane to provide a safe space for the nearby vehicle to cut into. With the successful cut-in maneuver completed, the host vehicle can smoothly follow the cut-in vehicle.

2.2. Potential Cut-In Vehicle Screening

In the cut-in scenario, a nearby vehicle performs a lane change operation from the original lane to in front of the host vehicle, which is considered as the cut-in behavior of the

nearby vehicle. If all vehicles around the host vehicle are used for cut-in recognition, the calculations of the ACC system will be increased. Therefore, it makes sense that vehicles in the two adjacent lanes are considered as potential cut-in vehicles.

Using the width of a city lane as a reference and the center of the vehicle directly in front of the host as the base point, the maximum lateral distance and minimum lateral distance of the area where the next vehicle exists is determined, as shown in Figure 3.

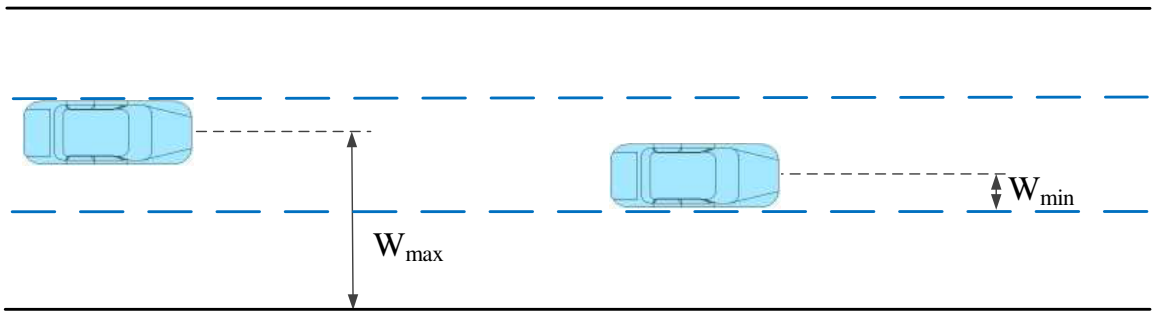

Figure 3. Determination of the lateral distance range.

W_{max} and W_{min} can be written as:

$$W_{\max} = 2H - \frac{1}{2}W_{ego} \tag{1}$$

$$W_{min} = \frac{1}{2}W_{ego} \tag{2}$$

where H is the width of the lane and W_{ego} is the width of the host vehicle.

By calculating the lateral distance of surrounding vehicles relative to the host vehicle, the vehicle of which the lateral distance meets the interval $[W_{\min}, W_{\max}]$ is regarded as the adjacent lane vehicle. As shown in Figure 4, nearby vehicles A and B are in the forward direction of the host vehicle, and the azimuth angles of the nearby vehicles are α and β, respectively. d_{relA} and d_{relB} are the relative distances between the nearby vehicles and the host vehicle. Therefore, the relative lateral distances d_{relA} and d_{relB} between the nearby vehicles and the host vehicle are expressed as follows.

$$d_{latA} = d_{relA} \sin\alpha \tag{3}$$

$$d_{latB} = d_{relB} \sin\beta \tag{4}$$

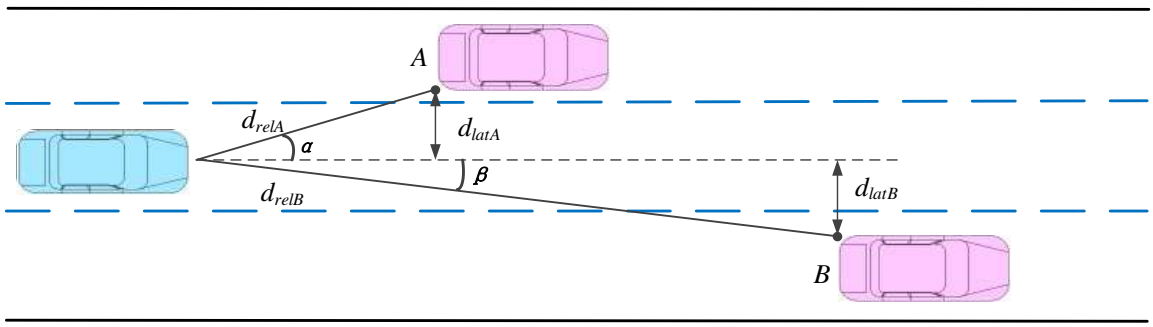

Figure 4. Calculation of the relative lateral distances of the nearby vehicles.

When the lateral distances, d_{relA} and d_{relB}, of nearby vehicles A and B meet the interval $[W_{\min}, W_{\max}]$, the nearby vehicles are both regarded as adjacent lane vehicles.

In the cut-in scenario, in addition to the relative lateral distance of the surrounding vehicles, the relative longitudinal distance of the next vehicle also needs to be considered.

Only the cut-in behavior of vehicles on both sides that are closest to the host vehicle in the longitudinal direction affect the driving state of the host vehicle. Therefore, the vehicles on both sides are selected as potential cut-in vehicles that are closest to the host vehicle in the longitudinal direction. As shown in Figure 5, H is the host vehicle. Assume that there are four target vehicles P(1), P(2), P (3), and P(4) in front of host vehicle H. P(1) is the follow target of host vehicle H. Since P(2) and P(3) are in the left and right lanes with the smallest relative longitudinal distance to the host vehicle, P(2) and P(3) are regarded as potential cut-in vehicles.

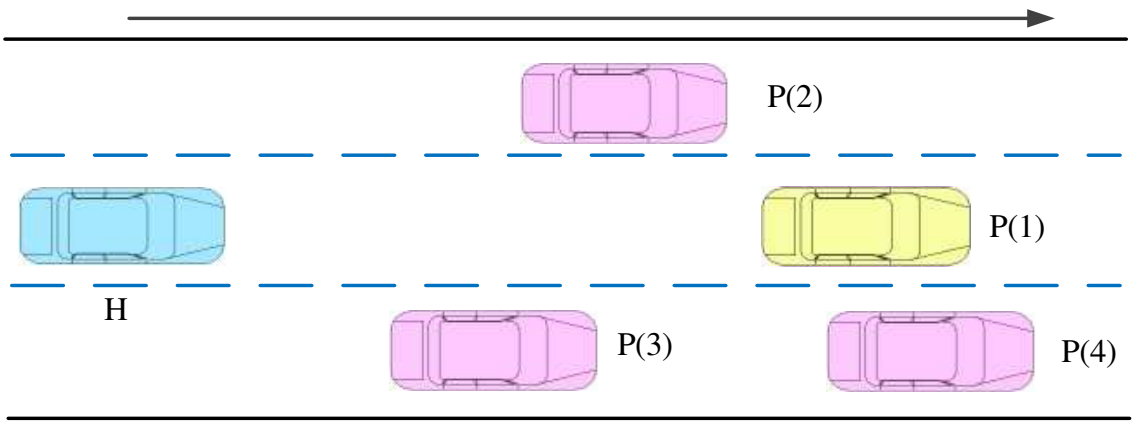

Figure 5. Determination of potential cut-in vehicles.

2.3. Quantification of Cut-In Possibility Based on Lateral Distance and Lateral Relative Velocity

When a nearby vehicle cuts into the lane, the lateral distance and lateral velocity of the vehicle change quite obviously [25]. Therefore, this paper studies the cut-in possibility of nearby vehicles in the lateral distance and with a relative lateral velocity.

2.3.1. Lateral Distance Criterion

When a nearby vehicle performs a cut-in maneuver, the continuous change of the nearby vehicle's lateral displacement is the most intuitive. The driver's judgment of the cut-in vehicle is generally based on whether the lateral distance of the nearby vehicle has an impact on the current driving trajectory of the host vehicle. As shown in Figure 6, if the nearby vehicle's body and the host vehicle's body overlap in the longitudinal direction, the driver will take measures to adjust the vehicle, according to the driving state of the two vehicles. Therefore, cut-in possibility is determined according to a change in the vehicle's lateral distance.

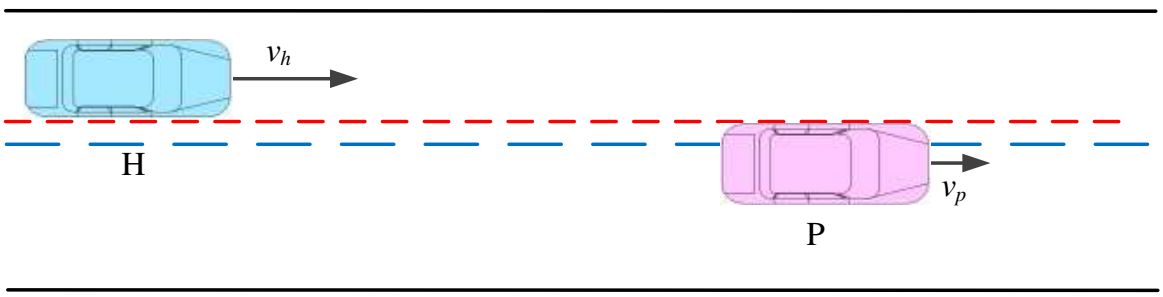

Figure 6. Vehicle's body overlap.

As the lateral displacement of a nearby car increases, the cut-in possibility of the vehicle becomes greater. The cut-in possibility is the greatest when the body of the nearby vehicle overlaps with the body of the host vehicle in the longitudinal direction. Therefore,

the cut-in possibility of the nearby vehicle is quantified as the value of [0,1] according to the vehicle's lateral distance.

$$P_{cutin}_d_{lat} \in [0,1] \tag{5}$$

where $P_{cutin}_d_{lat}$ is the cut-in possibility based on lateral distance. When $P_{cutin}_d_{lat} = 1$, it is considered that the nearby vehicle will definitely cut in. On the contrary, when $P_{cutin}_d_{lat} = 0$, it is considered that the nearby vehicle cannot cut in. In addition, because the lateral distance of the nearby vehicle is positively related to the cut-in possibility, the sigmoid function is selected to link the lateral displacement with the cut-in possibility.

$$P_{cutin}_d_{lat} = \begin{cases} \frac{1}{1+e^{-d_{lat}+a}} & d_{lat} \in [-H,0] \\ \frac{1}{1+e^{d_{lat}+b}} & d_{lat} \in [0,H] \end{cases} \tag{6}$$

where d_{lat} is the lateral distance of the nearby vehicle, a and b are both parameters that are related to the width of the lane line and the width of the host vehicle.

As the relative lateral distance decreases, the cut-in possibility of the nearby vehicle increases gradually. When the nearby vehicle body and the host vehicle body have any overlap in the longitudinal direction, $P_{cutin}_d_{lat} = 1$, and is regarded as cut-in behavior. The cut-in possibility based on lateral distance $P_{cutin}_d_{lat}$ and lateral distance d_{lat} are shown in Figure 7.

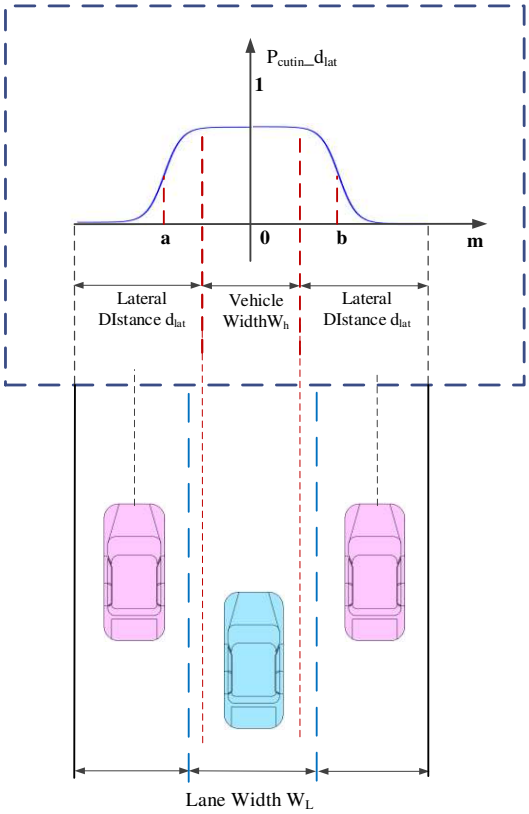

Figure 7. Cut-in possibility based on relative lateral distance.

As shown in Figure 7, the ordinate is the cut-in possibility based on lateral distance $P_{cutin}_d_{lat}$, the abscissa is the lateral distance of nearby vehicle, d_{lat}, W_{ego} is the width of the vehicle, and H is the width of the lane. As shown in Figure 7a, with the lateral distance of a nearby vehicle decreases, the cut-in possibility, $P_{cutin}_d_{lat}$, becomes greater. When a nearby vehicle shifts laterally until it overlaps with the host vehicle body in the longitudinal direction and touches the red dotted line in Figure 7a, the cut-in probability is considered to be 1, which means $P_{cutin}_d_{lat} = 1$ when $d_{lat} \leq (1/2)W_h$.

2.3.2. Relative Lateral Velocity Criterion

There can be a minor misjudgment in recognizing cut-in behavior only using the lateral distance. As shown in Figure 8, in the curve scenarios, for the cut-in vehicle, if the cut-in recognition is only performed using lateral distance, the turning behavior of the nearby car will be recognized as a cut-in behavior, which is obviously unreasonable. Therefore, the relative lateral velocity of the nearby car is used as another reference factor.

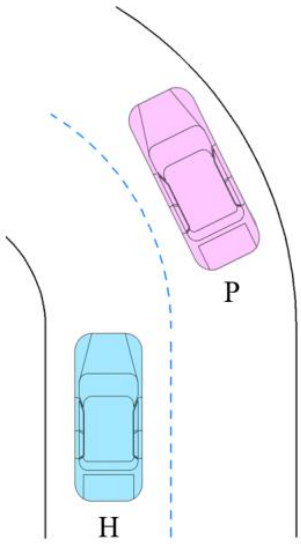

Figure 8. Curve scenarios.

During the cut-in process of a nearby vehicle, similar to the lateral distance, the lateral speed of the nearby vehicle also gradually increases, and then gradually decreases until the cut-in maneuver is successful [26]. At the same time, in curve scenarios, a nearby vehicle also has a large lateral speed. Therefore, in order to avoid judging the turning behavior of a vehicle as a cut-in behavior, this paper uses the relative lateral speed between the host vehicle and the nearby vehicle as a reference.

$$v_{rellat} = v_{plat} - v_{hlat} \tag{7}$$

where v_{rellat} is the relative lateral velocity, v_{plat} is the relative lateral velocity of the nearby vehicle, and v_{hlat} is the lateral velocity of the host vehicle.

Similar to the lateral distance judgment method, as shown in Figure 9, the sigmoid function is used to fit the relationship between cut-in probability and the relative lateral distance.

$$P_{cutin_}v_{rellat} = \frac{1}{1 + e^{v_{rellat} - c}} \tag{8}$$

where $P_{cutin_}v_{rellat}$ is the cut-in possibility of the lateral relative velocity and c is the parameter.

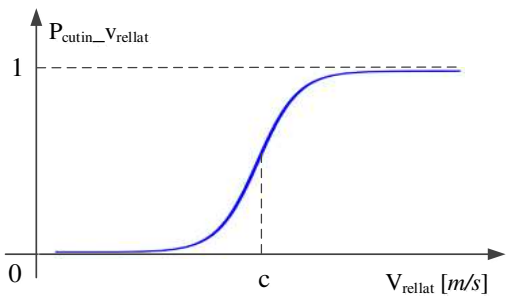

Figure 9. Cut-in possibility of lateral velocity.

In a straight lane scenario, as a nearby vehicle cuts in, the lateral velocity of the vehicle gradually increases, but the lateral velocity of the host vehicle is 0, so the relative lateral velocity between the host car and the nearby vehicle gradually increases, and the cut-in possibility becomes greater. In a curve scenario, both the host vehicle and the nearby vehicle have large lateral speeds, so the relative lateral velocity is small, which means that the cut-in possibility is low and the situation where a nearby vehicle that is driving normally on a curve is judged as a cut-in vehicle is avoided. If the nearby vehicle performs a cut-in maneuver, the vehicle needs to generate a greater lateral velocity than the host vehicle to cut in the front of the host vehicle successfully. Therefore, the relative lateral velocity criterion is effective when judging the cut-in possibility.

2.3.3. Quantification of Cut-In Possibilities

For the quantification of the cut-in possibility of vehicles on both sides, lateral distance $P_{cutin}_d_{lat}$ and lateral relative velocity $P_{cutin}_v_{rellat}$ are comprehensively considered. Because the lateral distance and lateral relative velocity have different influences on the cut-in possibility of a nearby vehicle under different road scenarios, the lateral distance $P_{cutin}_d_{lat}$ and the lateral relative velocity $P_{cutin}_v_{rellat}$ are set with different weights to adapt to the quantification of the cut-in possibility in different road scenes, so the cut-in possibility of the cut-in vehicle is finally expressed as

$$P_{cutin} = w_1 P_{cutin}_d_{lat} \cdot w_2 P_{cutin}_v_{rellat} \quad w_1, w_2 \in [0, 1] \tag{9}$$

where P_{cutin} is the cut-in possibility of the side car, w_1, w_2 is the weight.

Figure 10 shows the quantified cut-in possibility of nearby vehicles under different weights. The maximum value of cut-in probability P_{cutin} is 1, and the extreme value changes with different weights, so that the road scenarios factor is taken into account in the quantification of the cut-in behavior of a nearby vehicle, which avoids the wrong estimation of cut-in probability when facing changing road scenarios.

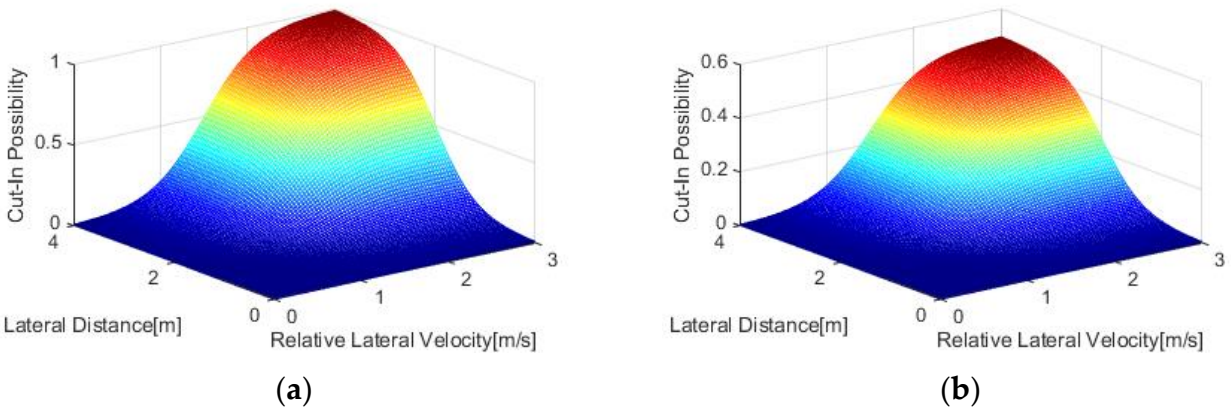

Figure 10. The cut-in possibility of under different weights: (**a**) the cut-in possibility under the weights of a straight lane. (**b**) The cut-in possibility under the weights for a curved lane.

In cut-in scenarios, as a nearby vehicle gradually cuts in, the cut-in possibility also changes at any time. In the host vehicle control, the generated cut-in probability value is used as a reference variable for the MPC–ACC, so that it can control the state of the host vehicle in a coordinated way to adapt to the cut-in vehicle according to the changing cut-in possibility. Finally, under the condition that the nearby vehicle safely cuts in, the host vehicle can achieve a smooth target conversion process.

3. The Cut-In Vehicle Safety Distance Model

In daily driving, facing cut-in behavior from the next car, the decision of the host vehicle will be directly affected by relative distance d_{rel} between the host vehicle and the preceding vehicle. On the one hand, when d_{rel} is large enough, as shown in Figure 11, there is enough cut-in space for the nearby vehicle and so the cut-in maneuver of the nearby vehicle will not affect the host vehicle; the host vehicle does not need to brake in advance, and instead switches the following target after the nearby vehicle cuts in. On the other hand, when d_{rel} is small, as shown in Figure 12, as the cut-in possibility becomes higher, and in order to ensure driving safety before a nearby vehicle cuts in, necessary braking measures must be taken to provide enough space for the nearby vehicle to cut in smoothly. Therefore, it is necessary to determine a minimum safe distance value to judge whether the cut-in a the nearby vehicle will affect the normal driving of the host vehicle.

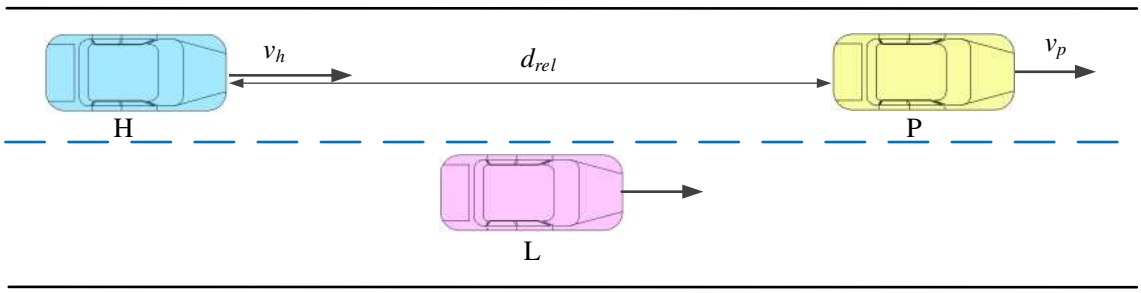

Figure 11. Sufficient relative distance.

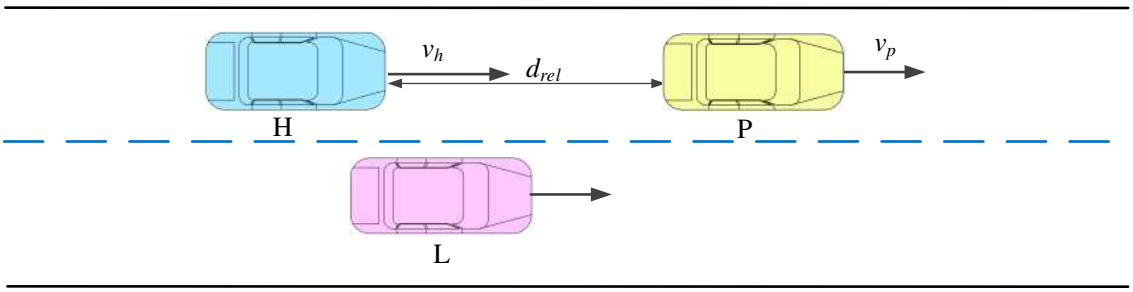

Figure 12. Insufficient relative distance.

3.1. The Safe Distance Model of the Cut-In Vehicle for a Straight Lane

Figure 13 shows a schematic diagram of a straight lane cut-in scenario. The cut-in vehicle, L, changes from the current lane to the target lane between vehicles H and P. Vehicles L, H, and P, respectively, represent the cut-in vehicle, the host vehicle, and the preceding vehicle.

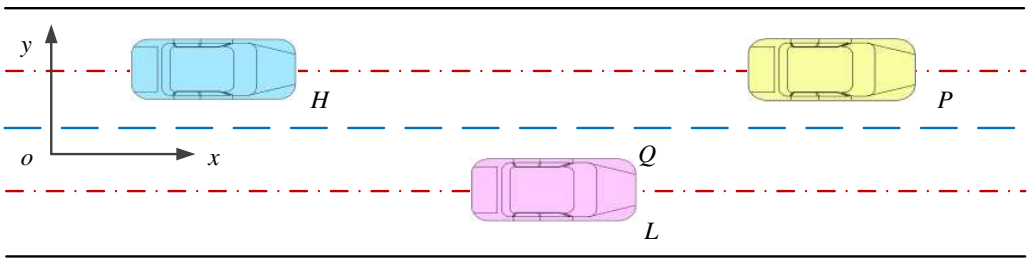

Figure 13. Straight lane cut-in scenario.

Suppose the start time when cut-in vehicle L executes the cut-in operation is $t = 0$. The lane change operation consists of two parts: the first part is the initialization phase,

in which cut-in vehicle L successfully spends time t_{abj} for the lane change to adjust the longitudinal distance relative to other vehicles that are around it, and its own longitudinal speed. The second part is to perform the steering operation to cut into the target lane. t_{abj} represents the adjustment time of the longitudinal distance and longitudinal speed required by the cut-in vehicle to successfully execute the lane change before starting the lane change operation.

In order to conveniently express the longitudinal and lateral distance among the involved vehicles, suppose that the geodetic coordinate system is as shown in Figure 13. O is the origin, the X axis points to the direction of the vehicle, and the Y axis that is perpendicular to the X axis points to the target lane. Therefore, the longitudinal acceleration, longitudinal velocity, longitudinal position, and lateral position are expressed as $a_i(t), v_i(t), x_i(t)$ and $y_i(t)$, where $i \in \{H, L, P\}$. To be more precise, the longitudinal distance and the lateral distance, respectively, represent the distance from the point at the upper left corner of every vehicle to the origin of the coordinate system.

3.1.1. The Minimum Longitudinal Safety Distance between Cut-In Vehicle L and Preceding Vehicle P

As shown in Figure 14, considering a critical collision scenario, assuming that cut-in vehicle L collides with preceding vehicle P in the target lane at point c, S represents the initial lateral distance between the upper edge of cut-in vehicle L and the lower edge of] preceding vehicle P. D represents the lane width.

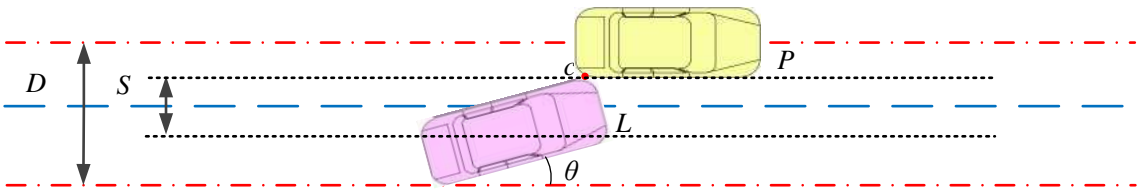

Figure 14. The position of cut-in vehicle L and preceding vehicle P.

The lateral acceleration of preceding vehicle P is 0, and the lateral position is constant. Assuming that the time at collision point c is $t_c + t_{adj}$, the collision avoidance condition between cut-in vehicle L and preceding vehicle P is given by the following formula:

$$x_L(t) < x_P(t) - l_0 - w \times \sin(\theta(t)) \quad \forall t \in \left[t_c + t_{adj}, T\right] \tag{10}$$

Where $x_L(t) x_P(t)$ is the abscissa of the cut-in vehicle and the preceding vehicle, respectively, l_0 is the length of the vehicles, and $\theta(t)$ is the yaw angle of the cut-in vehicle when changing lanes.

The last term in Formula (10) represents the distance between cut-in vehicle L and preceding vehicle P in the interval of $\left[t_c + t_{adj}, t_{lat} + t_{adj}\right]$ to avoid collision. The calculation is as follows:

$$\tan(\theta(t)) = \frac{v_{lat}(t)}{v_L(t)} \tag{11}$$

where $v_{lat}(t)$ is the lateral speed of the cut-in vehicle and $v_L(t)$ is the longitudinal speed of the cut-in vehicle. The largest $\theta(t)$, that is the largest $sin(\theta(t))$, is obtained at the $t = t_c + t_{adj}$ moment. Defining $L_P = l_0 + W \times sin(\theta(t_c + t_{adj}))$, the above formula is simplified to

$$x_L(t) < x_P(t) - L_P \quad \forall t \in \left[t_c + t_{adj}, T\right] \tag{12}$$

Use $S_r(t)$ to represent the longitudinal distance between cut-in vehicle L and preceding vehicle P during the lane change.

$$S_r(t) = x_P(t) - L_P - x_L(t) \quad \forall t \in \left[t_c + t_{adj}, T\right] \tag{13}$$

In order to make cut-in vehicle L not collide with preceding vehicle P during the lane change process, the following equation needs to be satisfied

$$S_r(t) = \left(\begin{array}{l} S_r(0) + \int_0^t \int_0^\lambda (a_P(\tau) - a_L(\tau)) d\tau d\lambda + \\ (v_P(0) - v_L(0)) \times t \end{array} \right) > 0 \quad \forall t \in \left[t_c + t_{adj}, T\right] \tag{14}$$

where $S_r(0) = x_P(0) - l_0 - x_L(0)$. In order not to collide with preceding vehicle P, the longitudinal distance ($S_r(t)$) between cut-in vehicle L and preceding vehicle P should be as large as possible during the period of $\left[t_c + t_{adj}, T\right]$, and $S_r(t)_{\max}$ will be used as the minimum safe distance, $MSD(L, P)$, at which point cut-in vehicle L will not collide with preceding vehicle P during the lane change process, thus, $MSD(L, P)$ is obtained:

$$MSD(L, P) = \underset{t}{Max} \left(\begin{array}{l} \int_0^t \int_0^\lambda (a_L(\tau) - a_P(\tau)) d\tau d\lambda + \\ (v_L(0) - v_P(0)) \times t + l_0 \end{array} \right) \quad \forall t \in \left[t_c + t_{adj}, T\right] \tag{15}$$

where a_L is the longitudinal acceleration of the cut-in vehicle, a_P is the longitudinal acceleration of the preceding vehicle, v_L is the longitudinal velocity of the cut-in vehicle, and v_P is the longitudinal velocity of the preceding vehicle

3.1.2. The Minimum Longitudinal Safety Distance between Cut-In Vehicle L and Host Vehicle H

As shown in Figure 15, considering the critical collision scenario, assuming that cut-in vehicle L collides with host vehicle H at point c, S represents the initial lateral distance between the upper edge of cut-in vehicle L and the lower edge of host vehicle H. D represents the lane width.

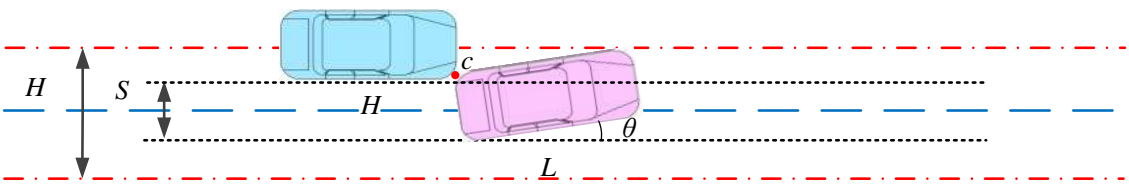

Figure 15. The position of cut-in vehicle L and host vehicle H.

The lateral acceleration of host vehicle H is 0 and the lateral position is constant. Assuming that the moment of collision, point c, is $t_c + t_{adj}$, in order to avoid a collision between cut-in vehicle L and host vehicle H, the following equation needs to be satisfied:

$$x_H(t) < x_L(t) - l_0 \times \cos(\theta(t)) \quad \forall t \in \left[t_c + t_{adj}, T\right] \tag{16}$$

where $x_P(t)$ is the abscissa of the host vehicle and l_0 is the length of vehicle.

Ignoring the influence of vehicle yaw, the above formula can be simplified to the following formula:

$$x_H(t) < x_L(t) - l_0 \quad \forall t \in \left[t_c + t_{adj}, T\right] \tag{17}$$

Use $S_r(t)$ to represent the longitudinal distance between cut-in vehicle L and host vehicle H during the lane change:

$$S_r(t) = x_L(t) - L - x_H(t) \quad \forall t \in \left[t_c + t_{adj}, T\right] \tag{18}$$

In order to prevent cut-in vehicle L from colliding with host vehicle H during the lane change process, the following equation needs to be satisfied:

$$S_r(t) = \begin{pmatrix} S_r(0) + \int_0^t \int_0^\lambda (a_L(\tau) - a_H(\tau))d\tau d\lambda + \\ (v_L(0) - v_H(0)) \times t \end{pmatrix} > 0 \quad \forall t \in \left[t_c + t_{adj}, T\right] \tag{19}$$

where $S_r(0) = x_L(0) - l_0 - x_H(0)$. The minimum safe distance, $MSD(L, H)$, at which cut-in vehicle L will not collide with host vehicle P during the lane change process is obtained

$$MSD(L, H) = \underset{t}{Max}\begin{pmatrix} \int_0^t \int_0^\lambda (a_H(\tau) - a_L(\tau))d\tau d\lambda \\ +(v_H(0) - v_L(0)) \times t + l_0 \end{pmatrix} \quad \forall t \in \left[t_c + t_{adj}, T\right] \tag{20}$$

where a_H is the longitudinal acceleration of the host vehicle and v_H is the longitudinal velocity of the host vehicle.

3.2. The Safe Distance Model of the Cut-In Vehicle for Curve Lane

The curved lane cut-in scenario is shown in Figure 16. Cut-in vehicle L changes from the current lane to the target lane between host vehicle H and preceding vehicle P. Vehicles H, L, and P, respectively, represent the host vehicle, the cut-in vehicle, and the preceding vehicle in the own lanes. Suppose that the outer lane and the inner lane have the same instantaneous center, the radius of curvature of the outer lane is R, and the lane width is D.

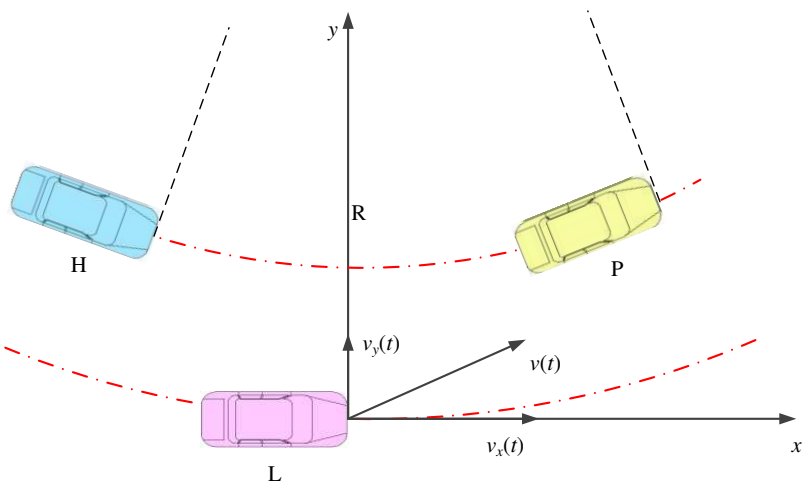

Figure 16. The position between the cut-in vehicle and the surrounding vehicles in a curved lane.

In order to express the longitudinal and lateral distance relationships between the involved vehicles, suppose that the geodetic coordinate system is as shown in Figure 16; O is the origin, the X axis points to the direction of the vehicle, and the Y axis that is perpendicular to the X axis points to the target lane. Therefore, longitudinal acceleration, longitudinal velocity, longitudinal position, and lateral position are expressed as $a_i(t), v_i(t), x_i(t)$ and $y_i(t)$, respectively, where $i \in \{H, L, P\}$. Similarly, the longitudinal distance and the lateral distance, respectively, represent the distance from the point at the upper left corner of every vehicle to the origin of the coordinate system.

3.2.1. The Minimum Longitudinal Safety Distance between Cut-In Vehicle L and Preceding Vehicle P

As shown in Figure 17 below, cut-in vehicle L changes lanes from the outer lane to the inner lane. The dashed-line vehicles are the positions of the two vehicles at the initial time, and the solid-line vehicles are the positions of the two vehicles at the possible collision time. The arc length from cut-in vehicle L along the outer lane to preceding vehicle P at the initial moment is $l_1(0)$. The arc length from preceding vehicle P along the inner lane to cut-in vehicle L at the initial moment is $l_2(0)$. The minimum safe distance between the host vehicle and the preceding vehicle at the initial moment is assumed to be $MSD(L, P)$.

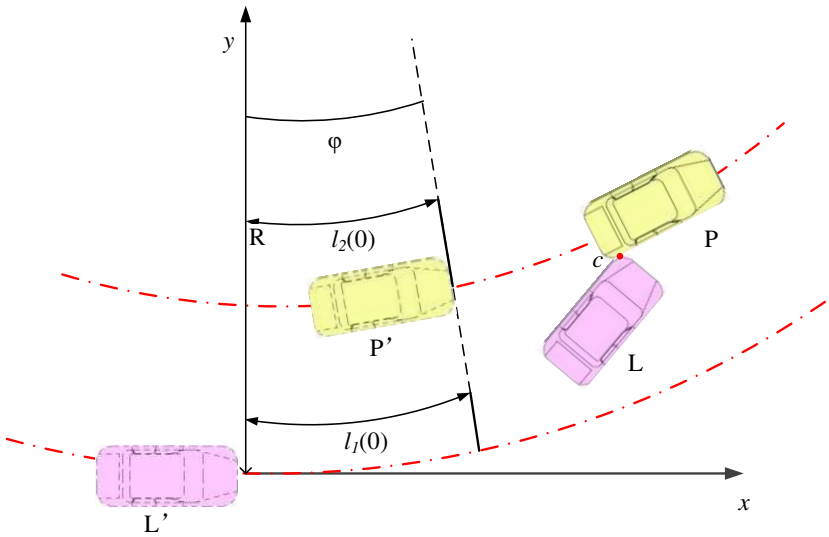

Figure 17. The positions of cut-in vehicle L and preceding vehicle P.

In order to make it so that cut-in vehicle L does not collide with preceding vehicle P during the lane change, the following equation needs to be satisfied:

$$S_r(t) = \left(\begin{array}{c} l_2(0) + \int_0^t \int_0^\lambda (a_P(\tau) - a_L(\tau)) d\tau d\lambda + \\ (v_P(0) - v_L(0)) \times t - l_0 \end{array} \right) > 0 \quad \forall t \in \left[t_c + t_{adj}, T \right] \tag{21}$$

According to the geometric relationship, $l_1(0)$ and $l_2(0)$ satisfy the following formula:

$$l_2(0) = \frac{l_1(0)}{R}(R - D) \tag{22}$$

The minimum safe distance between cut-in vehicle L and the preceding vehicle along the inner lane during the lane change is:

$$l_2(0) = Max\left[\int_0^t \int_0^\lambda \Delta a_{L-P} d\tau d\lambda + \Delta v_{L-P}(0) \times t + l_0 \right] \tag{23}$$

According to the law of cosines, the minimum safe distance, $MSD(L, P)$, between cut-in vehicle L and preceding vehicle P at the initial moment of lane change is:

$$MSD(L, P) = \begin{cases} \left(R^2 + (R - D)^2 - 2R(R - D)\cos\varphi \right)^{1/2}, & l_2(0) > 0 \\ -\left(R^2 + (R - D)^2 - 2R(R - D)\cos\varphi \right)^{1/2}, & l_2(0) < 0 \end{cases} \tag{24}$$

where φ is the central angle corresponding to $l_2(0)$, which is obtained as:

$$\varphi = \frac{l_2(0)}{R - D} \tag{25}$$

3.2.2. The Minimum Longitudinal Safe Distance between Cut-in Vehicle L and Host Vehicle H

As shown in Figure 18 below. Cut-in vehicle L changes lanes from the outer lane to the inner lane. The dashed-line vehicles are the positions of the two vehicles at the initial time, and the solid-line vehicles are the positions of the two vehicles at the possible collision time. The arc length from the cut-in vehicle along the outer lane to the host vehicle at the initial moment is $l_1(0)$, arc length from host vehicle H along the inner lane to cut-in vehicle L at the initial moment is $l_2(0)$, and the minimum safe distance between the cut-in vehicle L and host vehicle H at the initial moment is assumed to be $MSD(L, H)$.

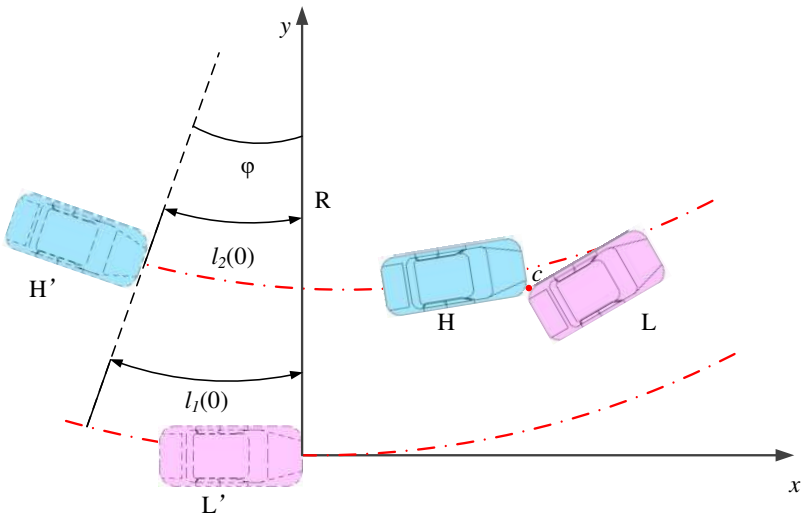

Figure 18. The positions of cut-in vehicle L and host vehicle H in the curved lane.

In order to make it so that cut-in vehicle L does not collide with host vehicle H during the lane change process, the following equation needs to be satisfied:

$$S_r(t) = \left(\begin{array}{c} l_2(0) + \int_0^t \int_0^\lambda (a_L(\tau) - a_H(\tau)) d\tau d\lambda + \\ (v_L(0) - v_H(0)) \times t - l_0 \end{array} \right) > 0 \quad \forall t \in \left[t_c + t_{adj}, T \right] \tag{26}$$

According to the geometric relationship, the calculation formula of $l_1(0)$ and $l_2(0)$ is:

$$l_2(0) = \frac{l_1(0)}{R}(R - D) \tag{27}$$

During the lane change process, the minimum safe distance between the cut-in vehicle and the host vehicle along the inner lane is:

$$l_2(0) = Max\left[\int_0^t \int_0^\lambda \Delta a_{H-L} d\tau d\lambda + \Delta v_{H-L}(0) \times t + l_0 \right] \tag{28}$$

According to the formula of the law of cosines, the minimum safe distance $MSD(L, H)$ between cut-in vehicle L and host vehicle H at the initial moment of lane change is:

$$MSD(L, H) = \begin{cases} \left(R^2 + (R - D)^2 - 2R(R - D)\cos\varphi \right)^{1/2}, & l_2(0) > 0 \\ -\left(R^2 + (R - D)^2 - 2R(R - D)\cos\varphi \right)^{1/2}, & l_2(0) < 0 \end{cases} \tag{29}$$

where φ is the central angle corresponding to $l_2(0)$, which is obtained using the following equation:

$$\varphi = \frac{l_2(0)}{R - D} \tag{30}$$

4. Host Vehicle Trajectory Tracking Control

This section uses the model predictive control (MPC) method to follow the trajectory of a cut-in vehicle and establishes a MPC-based ACC control system (MPC–ACC). After judging a vehicle's cut-in intention, the cut-in vehicle safety distance model is compared with the host vehicle and the preceding vehicle in front of host vehicle in the same lane. If the current relative distance meets the safety distance model, the cut-in vehicle safety distance model is used. The longitudinal trajectory of the cut-in vehicle after entering the lane is the reference trajectory; if the current relative distance does not meet the safety distance model, the cut-in vehicle's lateral trajectory and longitudinal trajectory are used as the reference trajectory at the same time to ensure that the host vehicle can be brake in advance and follow the cut-in vehicle to drive smoothly.

This section uses the model predictive control (MPC) method to follow the trajectory of the cut-in vehicle and establishes an ACC control system based on MPC (MPC–ACC). After screening the potential cut-in vehicles, the safe distance model of the cut-in vehicle is compared with the relative distance between the host vehicle and the preceding vehicle in front of it in the lane. If the current relative distance meets the safe distance model, the longitudinal trajectory of the cut-in vehicle is taken as the reference trajectory of the MPC–ACC. If the current relative distance does not meet the safe distance model, the cut-in possibility and the longitudinal trajectory of the cut-in vehicle are used as the reference trajectory of the MPC–ACC to ensure that the vehicle can gradually decelerate as the cut-in possibility of the nearby vehicle increases so that enough space can be provided for the cut-in vehicle to successfully change lanes. At the same time, constraints were set in the vehicle trajectory tracking control to ensure driver safety and comfort.

4.1. Inter-Vehicle Longitudinal Kinematics Model

In this paper, constant time headway (CTH) is adopted for the MPC–ACC. In the CTH strategy, the time headway is a fixed value, and the inter-vehicle distance is proportional to the host vehicle's speed. In order to ensure safety, the minimum safe distance is considered in the strategy, as shown in the following equation:

$$\Delta x_{des} = t_h v_H + x_0 \tag{31}$$

where Δx_{des} is the desired inter-vehicle distance, t_h is the time headway and x_0 is minimum safe distance added to ensure safety, which is constant.

When the host vehicle follows the preceding vehicle in a stable state, the trajectory of the preceding vehicle needs to be used as the reference trajectory of the MPC–ACC. At the same time, to consider the safety of following, the reference trajectory of the host vehicle is obtained:

$$x_{ref}(k) = x_H(k) - \Delta x_{des} \tag{32}$$

where $x_{ref}(k)$ is the longitudinal reference trajectory of the host vehicle and $x_H(k)$ is the host vehicle's coordinates.

According to the inter-vehicle longitudinal kinematics, as shown in Figure 19, the longitudinal kinematics model of the vehicles can be obtained, as shown using the following equations

$$\Delta x(k+1) = \Delta x(k) + v_{rel}(k)T_s + \frac{1}{2}(a_P(k) - a_H(k))T_s^2 \tag{33}$$

$$v_{rel}(k+1) = v_{rel}(k) + [a_P(k) - a_H(k)]T_s \tag{34}$$

$$v_H(k+1) = v_H(k) + a_H(k)T_s \tag{35}$$

where $a_H(k)$, $a_P(k)$ is acceleration of the host vehicle and the preceding vehicle, respectively, $v_{rel}(k)$ is the relative velocity, which satisfies the equation $v_{rel}(k) = v_H(k) - v_P(k)$, where is v_H the host vehicle velocity and v_P is the preceding vehicle velocity; $\Delta x(k)$ is the relative distance, which satisfies $\Delta x(k) = x_P(k) - x_H(k)$, $x_P(k)$ is the preceding vehicle's coordinates, and T_s is the sampling time.

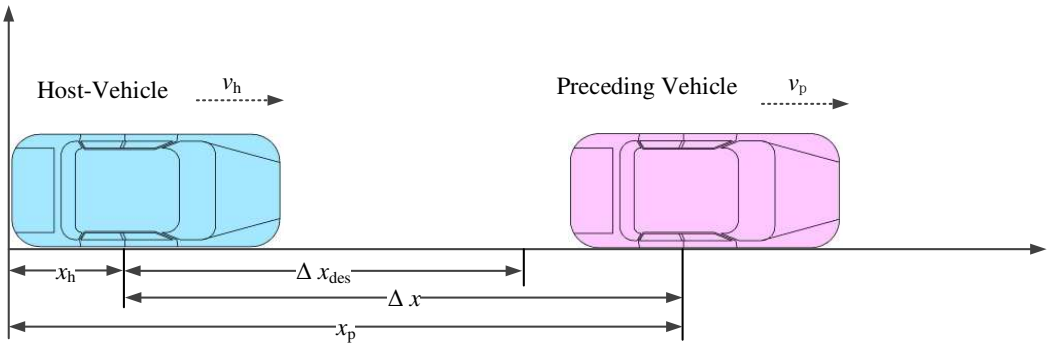

Figure 19. Vehicle longitudinal kinematics model.

In ACC, a hierarchical control is usually adopted. Because of the foundation of lower-level control, the vehicle cannot usually immediately reach a desired acceleration output using the upper-level control. Therefore, the following equations is used to reflect the delay characteristics of the acceleration control response [27].

$$a_H(k+1) = (1 - \frac{T_s}{\tau})a_H(k) + \frac{T_s}{\tau}u(k) \tag{36}$$

$$Jerk_P(k+1) = -\frac{1}{\tau}a_P(k) + \frac{1}{\tau}u(k) \tag{37}$$

where $u(k)$ is the desired acceleration output by the upper control system and $Jerk_H$ is the jerk of the host vehicle.

Select relative distance, relative vehicle velocity, host vehicle velocity, host vehicle acceleration, and the host vehicle jerk as the state quantities of the longitudinal kinematics model:

$$x(k) = [\Delta x(k), v_{rel}(k), v_H(k), a_H(k), Jerk_H(k)]^T \tag{38}$$

Select the relative distance and the relative velocity as the output vector of the longitudinal kinematics model:

$$y_c(k) = [\Delta x(k), v_{rel}(k)]^T \tag{39}$$

The acceleration of the preceding vehicle is regarded as a disturbance of the control system, so the state equation of the control system can be obtained:

$$\boldsymbol{x}(k+1) = \boldsymbol{A}\boldsymbol{x}(k) + \boldsymbol{B}u(k) + \boldsymbol{G}w(k) \tag{40a}$$

$$\boldsymbol{y}_c(k) = \boldsymbol{C}\boldsymbol{x}(k) + \boldsymbol{Z} \tag{40b}$$

where,

$$\boldsymbol{A} = \begin{bmatrix} 1 & T_s & 0 & -T_s^2/2 & 0 \\ 0 & 1 & 0 & -T_s & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 - T_s/\tau & 0 \\ 0 & 0 & 0 & -1/\tau & 0 \end{bmatrix}, \boldsymbol{B} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ T_s/\tau \\ 1/\tau \end{bmatrix}$$

$$\boldsymbol{G} = \begin{bmatrix} T_s^2/2 \\ T_s \\ T_s \\ 0 \\ 0 \end{bmatrix}, \boldsymbol{C} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}^T, \boldsymbol{Z} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Equations (40a) and (40b) is the longitudinal kinematics control model of the ACC system. Compared with the conventional two-degree-of-freedom vehicle model, this model considers the acceleration disturbance of the preceding vehicle and reproduces the dynamic evolution law of the entire system more truly and comprehensively. In addition,

by describing the acceleration and jerk of the host vehicle, the accuracy and reliability of the model are improved.

4.2. Reference Trajectory Generation

4.2.1. Vehicle Hollowing Scenarios

In driving scenario 1, because there is no cut-in vehicle, the host vehicle steadily follows the preceding vehicle. When the FSM is in this state, the relative distance between the host vehicle and the preceding vehicle converges to the desired inter-vehicle distance, and the relative vehicle velocity converges to 0, so the reference trajectory vector of the MPC–ACC can be obtained:

$$\boldsymbol{y}_{ref}(k) = [\Delta x_{des}, 0]^T \tag{41}$$

Substituting Equation (31) into Equation (41):

$$\boldsymbol{y}_{ref}(k) = \boldsymbol{C}_{ref} x(k) + \boldsymbol{Z}_{ref} \tag{42}$$

where,

$$\boldsymbol{C}_{ref} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ t_h & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}^T, \boldsymbol{Z}_{ref} = \begin{bmatrix} x_0 \\ 0 \end{bmatrix}$$

4.2.2. Cut In beyond Safe Distance

In driving scenario 2, although there is a cut-in vehicle, because the relative distance between the host vehicle and the preceding vehicle meets the safe distance model of the cut-in vehicle, the cut-in vehicle has enough space to cut in. Therefore, the host vehicle does not need to brake in advance, but only switches the following target from the preceding vehicle to the cut-in vehicle. The trajectory of the cut-in vehicle is:

$$x_L(k+1) = x_L(k) + v_L(k) T_s \tag{43}$$

where $x_L(k)$ is the cut-in vehicle longitudinal coordinate and v_L is the velocity of the cut-in vehicle.

The reference trajectory of the host vehicle is:

$$x_{ref}(k) = x_L(k) - x_{des}(k) \tag{44}$$

Therefore, state quantities $\Delta x(k)$, $v_{rel}(k)$ are rewritten as:

$$\Delta x(k) = x_L(k) - x_H(k) \tag{45}$$

$$\Delta x(k+1) = \Delta x(k) + v_{rel}(k) T_s + \frac{1}{2}(a_L(k) - a_H(k)) T_s^2 \tag{46}$$

$$v_{rel}(k) = v_L(k) - v_H(k) \tag{47}$$

$$v_{rel}(k+1) = v_{rel}(k) + [a_L(k) - a_H(k)] T_s \tag{48}$$

Similar to the vehicle following scenarios, the reference trajectory vector of the MPC–ACC in this scenario is:

$$\boldsymbol{y}_{ref}(k) = [\Delta x_{des}, 0]^T \tag{49}$$

$$\boldsymbol{y}_{ref}(k) = \boldsymbol{C}_{ref} x(k) + \boldsymbol{Z}_{ref} \tag{50}$$

4.2.3. Cut In with Safe Distance

In driving scenario 3, the relative distance between the host vehicle and the preceding vehicle does not meet the safe distance model of the cut-in vehicle. Therefore, the

host vehicle needs to gradually slow down as the cut-in possibility of the nearby vehicle increases, which expands the longitudinal relative distance to the preceding vehicle and provides enough space for the cut-in vehicle to cut in. In this process, the host vehicle needs to assign the cut-in vehicle as the following target. The longitudinal trajectory and the cut-in possibility of the nearby vehicle are considered as the reference for the MPC–ACC. Similarly, the state quantities are rewritten as:

$$\Delta x(k+1) = \Delta x(k) + v_{rel}(k)T_s + \frac{1}{2}(a_L(k) - a_H(k))T_s^2 \tag{51}$$

$$v_{rel}(k+1) = v_{rel}(k) + [a_L(k) - a_H(k)]T_s \tag{52}$$

Introducing cut-in possibility P_{cutin}, obtained in Section 2, into the reference, the relative distance between the host vehicle and the cut-in vehicle needs to be changed with cut-in possibility P_{cutin} so as to realize coordinated control of the host vehicle with the cut-in vehicle, so the reference trajectory of the relative distance $\Delta x(k)$ is:

$$\Delta x_{ref}(k) = [\Delta x_{des}(k) - \Delta x_0]P_{cutin} + \Delta x_0 \; 0 \le P_{cutin} \le 1 \tag{53}$$

where Δx_0 is the initial relative distance between the host vehicle and the cut-in vehicle, and P_{cutin} is the cut-in possibility. Reference trajectory $\Delta x_{ref}(k)$ changes with the cut-in possibility, as shown in Figure 20.

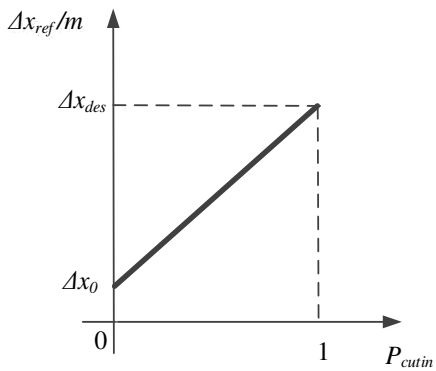

Figure 20. The law that the reference trajectory changes with the cut-in possibility.

Reference trajectory vector of the MPC–ACC system in driving scenario 3 is:

$$\boldsymbol{y}_{ref}(k) = [(\Delta x_{des}(k) - \Delta x_0)P_{cutin} + \Delta x_0, 0]^T \tag{54}$$

$$\boldsymbol{y}_{ref}(k) = \boldsymbol{C}_{ref}x(k) + \boldsymbol{Z}_{ref} \tag{55}$$

where,

$$\boldsymbol{C}_{ref} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ t_h P_{cutin} & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}^T, \boldsymbol{Z}_{ref} = \begin{bmatrix} x_0 P_{cutin} - \Delta x_0 P_{cutin} + \Delta x_0 \\ 0 \end{bmatrix}$$

4.3. Objective Function and Constraint Establishment

The control purpose of the ACC–MPC is that the host vehicle can stably follow the reference trajectory of the system under the constraints of ensuring safety and comfort, so the objective function is:

$$J = \sum_{i=1}^{p} \|\boldsymbol{Q}(\boldsymbol{y}_c(k+i|k) - \boldsymbol{y}_{ref}(k+i))\|^2 + \sum_{i=1}^{m-1} \|\boldsymbol{R}\boldsymbol{u}(k+i)\|^2 \tag{56}$$

Where $\boldsymbol{Q}$ and $\boldsymbol{R}$ are the weight coefficient matrices, respectively, $\boldsymbol{u}$ is the control vector. p is the prediction horizon, m is the control horizon, $p \geq m$, $\boldsymbol{y}_c(k+i|k)$ is the prediction output vector of time $k+i$ by the MPC–ACC system at k time, and $\boldsymbol{y}_{ref}(k+i)$ is the reference trajectory of the system at $k+i$ time. The penalty term $\boldsymbol{y}_c(k+i|k) - \boldsymbol{y}_{ref}(k+i)$ is to ensure that the difference between the output trajectory of the system and the reference trajectory is as small as possible; that is, during the following process, the relative distance between the host vehicle and the preceding vehicle or the cut-in vehicle is as close as possible to the inter-vehicle distance. The relative velocity between them is as close as possible to 0. The penalty term $\boldsymbol{u}(k+i)$ is to make the desired acceleration as small as possible.

When solving the optimal solution of the MPC–ACC, it is also necessary to satisfy various constraints in the process of following the preceding vehicle. First of all, in order to ensure the driving safety of the host vehicle during following, the longitudinal relative distance between the host vehicle and the preceding vehicle or the cut-in vehicle must meet the minimum safe distance constraint:

$$\Delta x(k) \geq d_c \tag{57}$$

where d_c is the minimum safety relative distance.

Secondly, considering the road velocity limit, the host vehicle's maximum acceleration capacity, maximum braking capacity, the velocity, and acceleration constraints are:

$$\begin{array}{c} v_{Hmin} \leq v_H(k) \leq v_{Hmax} \\ a_{Hmin} \leq a_H(k) \leq a_{Hmax} \\ u_{min} \leq u(k) \leq u_{max} \end{array} \tag{58}$$

In addition, driving comfort can be expressed by jerk. The smaller the absolute value of jerk is, the higher the driving comfort that can be obtained [28]. In order to ensure driving comfort, the host vehicle satisfies the constraints:

$$Jerk_{min} \leq Jerk_H(k) \leq Jerk_{max} \tag{59}$$

The final constraints are:

$$\begin{cases} \Delta x(k) \geq d_c \\ v_{Hmin} \leq v_H(k) \leq v_{Hmax} \\ a_{Hmin} \leq a_H(k) \leq a_{Hmax} \\ u_{min} \leq u(k) \leq u_{max} \\ Jerk_{min} \leq Jerk_H(k) \leq Jerk_{max} \end{cases} \tag{60}$$

Based on Equation (56) and inequalities (60), the optimization problem of the MPC–ACC can be formulated as follows

$$\min_{U(k)} J(x(k), u(k)) \tag{61}$$

subject to:

$$\Delta x(k) \geq d_c$$

$$v_{Hmin} \leq v_H(k) \leq v_{Hmax}$$

$$a_{Hmin} \leq a_H(k) \leq a_{Hmax}$$

$$u_{min} \leq u(k) \leq u_{max}$$

$$Jerk_{min} \leq Jerk_H(k) \leq Jerk_{max}$$

where,

$$J = \sum_{i=1}^{p} \|\boldsymbol{Q}(\boldsymbol{y}_c(k+i|k) - \boldsymbol{y}_{ref}(k+i))\|^2 + \sum_{i=1}^{m-1} \|\boldsymbol{R}\boldsymbol{u}(k+i)\|^2$$

4.4. Problem Solving

In order to determine the analytical solution of the problem, the MPC optimization problem needs to be transformed into a constrained quadratic program [29], as shown in the following formula:

$$\min_{U(k+m)} \left\{ \boldsymbol{U}(k+m)^T \boldsymbol{K}_1 \boldsymbol{U}(k+m) + 2\boldsymbol{K}_2 \boldsymbol{U}(k+m) \right\} \tag{62}$$

subject to:

$$\boldsymbol{\Omega} \boldsymbol{U}(k+m) \geq \boldsymbol{T} \tag{63}$$

where $\boldsymbol{K}_1$, $\boldsymbol{K}_2$, $\boldsymbol{\Omega}$, and $\boldsymbol{T}$ are coefficient matrices.

According to the principle of the MPC [30], the future behavior of the system is predicted according to the prediction model at each moment of system sampling, and the performance indicators of system in the future moment are optimized. The control sequence is obtained by solving the corresponding optimization problem and the first control quantity is applied to the system; then the prediction time domain is aligned one step forward. Finally, the process is repeated continuously. Therefore, according to the inter-vehicle longitudinal kinematics established in Section 4.1, the future behavior of the ACC system is predicted, as shown in the following equation:

$$\hat{\boldsymbol{X}}(k+p|k) = \bar{\boldsymbol{A}} x(k) + \bar{\boldsymbol{B}} \boldsymbol{U}(k+m) + \bar{\boldsymbol{G}} \boldsymbol{W}(k+p) \tag{64}$$

$$\hat{\boldsymbol{Y}}(k+p|k) = \bar{\boldsymbol{C}} x(k) + \bar{\boldsymbol{D}} \boldsymbol{U}(k+m) + \bar{\boldsymbol{E}} \boldsymbol{W}(k+p) + \bar{\boldsymbol{Z}} \tag{65}$$

where,

$$\hat{\boldsymbol{X}}(k+p|k) = \begin{bmatrix} \hat{\boldsymbol{x}}(k+1|k) \\ \hat{\boldsymbol{x}}(k+2|k) \\ \vdots \\ \hat{\boldsymbol{x}}(k+p|k) \end{bmatrix}, \hat{\boldsymbol{Y}}(k+p|k) = \begin{bmatrix} \hat{\boldsymbol{y}}_c(k+1|k) - \hat{\boldsymbol{y}}_{ref}(k+1|k) \\ \hat{\boldsymbol{y}}_c(k+2|k) - \hat{\boldsymbol{y}}_{ref}(k+1|k) \\ \vdots \\ \hat{\boldsymbol{y}}_c(k+p|k) - \hat{\boldsymbol{y}}_{ref}(k+1|k) \end{bmatrix}$$

$$\boldsymbol{U}(k+m) = \begin{bmatrix} u(k) \\ u(k+1) \\ \vdots \\ u(k+m-1) \end{bmatrix}, \boldsymbol{W}(k+p) = \begin{bmatrix} w(k) \\ w(k+1) \\ \vdots \\ w(k+p-1) \end{bmatrix}$$

where $\hat{\boldsymbol{x}}(k+1|k)$ is the system state prediction of time $k+1$ at k time; similarly, $\hat{\boldsymbol{y}}_c(k+1|k)$ is the prediction of the system output. The control sequence $u(k), u(k+1), \ldots, u(k+m-1)$ is the control variable to be solved. $w(k) = w(k+1) = \ldots = w(k+p-1)$ is the disturbance in the system; that is, the acceleration of the preceding vehicle, which is a constant value in the entire prediction time horizon. The prediction matrices in Equations (64) and (65) are:

$$\bar{\boldsymbol{A}} = \begin{bmatrix} \boldsymbol{A} \\ \boldsymbol{A}^2 \\ \vdots \\ \boldsymbol{A}^{p-1} \end{bmatrix}, \bar{\boldsymbol{B}} = \begin{bmatrix} \boldsymbol{B} & 0 & \cdots & 0 \\ \boldsymbol{AB} & \boldsymbol{B} & \cdots & \vdots \\ \cdots & \cdots & \vdots & 0 \\ \boldsymbol{A}^{p-1}\boldsymbol{B} & \boldsymbol{A}^{p-2}\boldsymbol{B} & \cdots & \sum_{i=0}^{p-m} \boldsymbol{A}^i \boldsymbol{B} \end{bmatrix},$$

$$\bar{G} = \begin{bmatrix} G & 0 & \cdots & 0 \\ AG & G & \cdots & \vdots \\ \cdots & \cdots & \vdots & 0 \\ A^{p-1}G & A^{p-2}G & \cdots & G \end{bmatrix}, \bar{C} = \begin{bmatrix} \left(C - C_{ref}\right)A \\ \left(C - C_{ref}\right)A^2 \\ \vdots \\ \left(C - C_{ref}\right)A^{p-1} \end{bmatrix},$$

$$\bar{D} = \begin{bmatrix} \left(C - C_{ref}\right)B & 0 & \cdots & 0 \\ \left(C - C_{ref}\right)AB & \left(C - C_{ref}\right)B & \cdots & \vdots \\ \cdots & \cdots & \vdots & 0 \\ \left(C - C_{ref}\right)A^{p-1}B & \left(C - C_{ref}\right)A^{p-2}B & \cdots & \sum_{i=0}^{p-m}\left(C - C_{ref}\right)A^{i}B \end{bmatrix},$$

$$\bar{E} = \begin{bmatrix} \left(C - C_{ref}\right)G & 0 & \cdots & 0 \\ \left(C - C_{ref}\right)AG & \left(C - C_{ref}\right)G & \cdots & \vdots \\ \cdots & \cdots & \vdots & 0 \\ \left(C - C_{ref}\right)A^{p-1}G & \left(C - C_{ref}\right)A^{p-2}G & \cdots & \left(C - C_{ref}\right)G \end{bmatrix}, \bar{Z} = \begin{bmatrix} Z - Z_{ref} \\ Z - Z_{ref} \\ \vdots \\ Z - Z_{ref} \end{bmatrix},$$

Substituting Equation (65) into Equation (56), and ignoring terms that have nothing to do with the control quantities:

$$\begin{aligned} J = {} & U(k+m)^T\left(\bar{D}^T\bar{Q}\bar{D} + \bar{R}\right)U(k+m) \\ & +2\left[x^T(k)\bar{C}^T\bar{Q}\bar{D} + W(k+p)^T\bar{E}^T\bar{Q}\bar{D} + \bar{Z}^T\bar{Q}\bar{D}\right]U(k+m) \end{aligned} \tag{66}$$

where,

$$\bar{Q} = \begin{bmatrix} Q & \cdots & 0 \\ \vdots & \vdots & \vdots \\ 0 & \cdots & Q \end{bmatrix} \bar{R} = \begin{bmatrix} R & \cdots & 0 \\ \vdots & \vdots & \vdots \\ 0 & \cdots & R \end{bmatrix}$$

Similarly, the system constraints should also be transformed. According to the constraint variables obtained from the previous discussion, the output equation of the constraint is:

$$y_b(k) = C_b x(k) \tag{67}$$

where,

$$C_b = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Substituting Equation (69) into the inequalities (61), the following equation is obtained:

$$\begin{cases} Y_{b\min} \le Y_b(k+p|k) \le Y_{b\max} \\ -U(k+m) \ge -U_{\max} \\ U(k+m) \ge U_{\min} \end{cases} \tag{68}$$

where,

$$Y_b(k+p|k) = \begin{bmatrix} y_b(k+1|k) \\ y_b(k+2|k) \\ \vdots \\ y_b(k+p|k) \end{bmatrix} = \bar{C}_b\hat{X}(k+p|k)$$

where,

$$\bar{C}_b = \begin{bmatrix} C_b & & & \\ & C_b & & \\ & & \ddots & \\ & & & C_b \end{bmatrix}$$

$$Y_{b\min} = \begin{bmatrix} y_{b\min} \\ y_{b\min} \\ \vdots \\ y_{b\min} \end{bmatrix} Y_{b\max} = \begin{bmatrix} y_{b\max} \\ y_{b\max} \\ \vdots \\ y_{b\max} \end{bmatrix}$$

$$y_{b\min} = \begin{bmatrix} d_c \\ v_{Hmin} \\ a_{Hmin} \\ Jerk_{min} \end{bmatrix} y_{b\max} = \begin{bmatrix} Inf \\ v_{Hmax} \\ a_{Hmax} \\ Jerk_{mxa} \end{bmatrix}$$

where Inf represents infinity.

$$\begin{bmatrix} -I \\ I \end{bmatrix} U(k+m) \geq \begin{bmatrix} -U_{\max} \\ U_{\min} \end{bmatrix} \tag{69}$$

where,

$$\Omega = \begin{bmatrix} -\bar{C}_b\bar{B} \\ \bar{C}_b\bar{B} \\ -I \\ I \end{bmatrix}, T = \begin{bmatrix} -Y_{b\max} + \bar{C}\bar{A}x(k) + \bar{C}\bar{G}W(k+p) \\ Y_{b\min} - \bar{C}\bar{A}x(k) - \bar{C}\bar{G}W(k+p) \\ -U_{\max} \\ U_{\min} \end{bmatrix}$$

Finally, the optimization problem of the MPC–ACC is turned into a quadratic programming problem:

$$\min_{U(k+m)} \left\{ U(k+m)^T K_1 U(k+m) + 2K_2 U(k+m) \right\}$$

subject to:

$$\Omega U(k+m) \geq T$$

where,

$$K_1 = \bar{D}^T\bar{Q}\bar{D} + \bar{R},\ K_2 = x^T(k)\bar{C}^T\bar{Q}\bar{D} + W(k+p)^T\bar{E}^T\bar{Q}\bar{D} + \bar{Z}^T\bar{Q}\bar{D}$$

By solving the quadratic programming problem, control vector $U(k+m)$ is obtained, and the first term $u(k)$ of the control vector $U(k+m)$ is applied to the system as the current control quantity.

5. Simulation Results and Analysis

The MPC–ACC for the cut-in scenarios was verified using the Simulink–Carsim ® platform. In this paper, different scenarios were adopted to verify the effectiveness of the proposed control strategy. In order to highlight the host vehicle's ability to brake before a nearby vehicle cut in, cut-in scenarios using a straight road under urban driving conditions was adopted in this paper. The complete vehicle model was built in Carsim ®.

5.1. Vehicle Following Scenario

In driving scenario 1, the vehicle in the nearby lane did not perform cut-in behavior. In the FSM, the host vehicle was in a following the preceding vehicle state at that time. The host vehicle used the longitudinal displacement and speed of the preceding vehicle as a reference to follow the vehicle stably, while also meeting the constraints of safety and driving comfort. The longitudinal velocity of the host vehicle, the relative distance from the

preceding vehicle, acceleration of the host vehicle, and jerk of the host vehicle are shown in Figure 21.

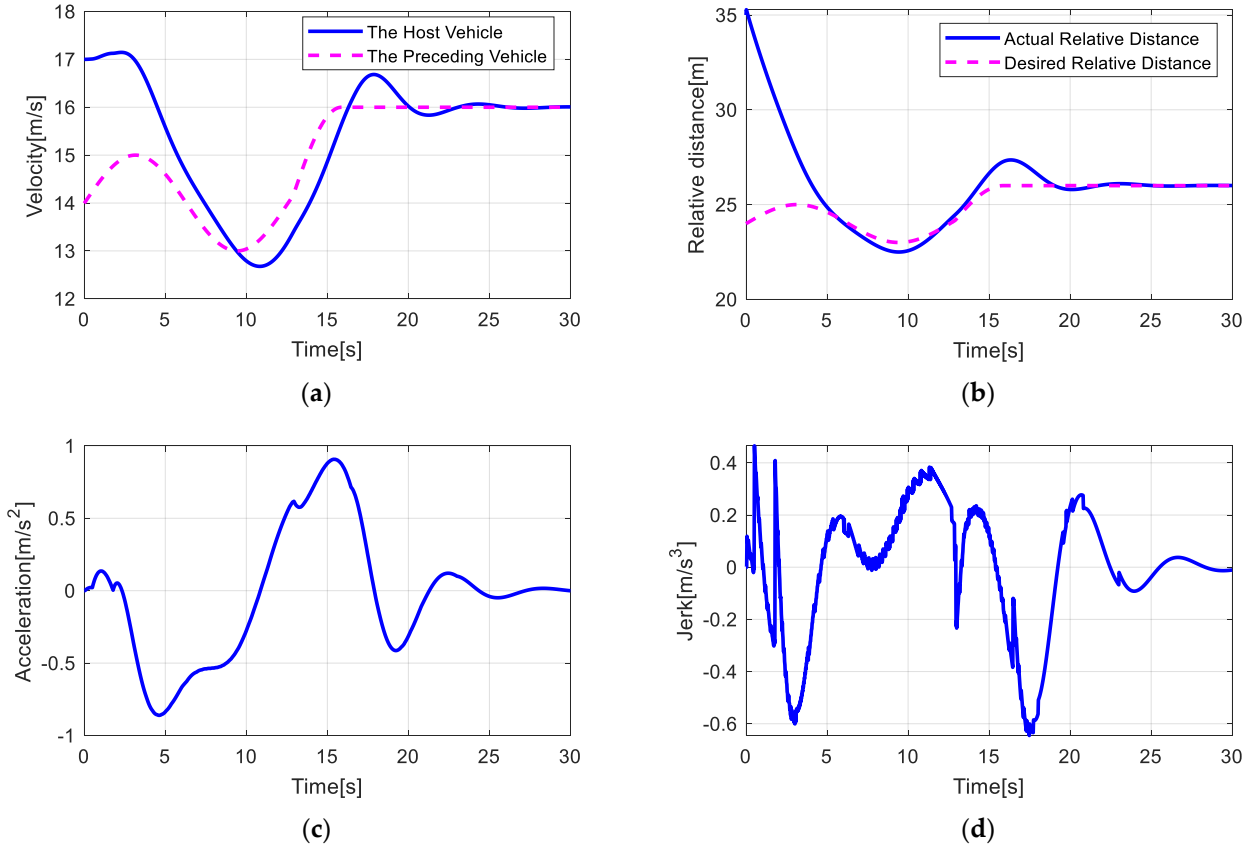

Figure 21. The simulation results of vehicle following scenarios: (**a**) the velocity of the host vehicle and the preceding vehicle; (**b**) the relative distance between the host vehicle and the preceding vehicle; (**c**) the acceleration of the host vehicle; and (**d**) the jerk of the host vehicle.

In Figure 21, the host vehicle can use the MPC–ACC control system to stably follow the preceding vehicle. Since there was no the cut-in vehicle, the host vehicle adopted the corresponding speed after recognizing the presence of the preceding vehicle to maintain its speed consistent with that of the vehicle and to maintain a safe relative distance. In Figure 21a, the initial speed of the vehicle was 17 m/s. After the presence of the leading vehicle was detected at time 0, the host vehicle began to decelerate, and gradually coincided with the speed of the preceding vehicle. The relative distance between this host vehicle and the preceding vehicle, as shown in Figure 21b, gradually decreased from the initial 35 m to 24 m, which met the conditions of driving safety. When the vehicle was close to the preceding vehicle, the acceleration always changed at ± 1.5 and jerk at ± 1, as shown in Figure 21c,d and Table 1. Because the acceleration and jerk change range is small, driving comfort was guaranteed.

Table 1. Statistics of the acceleration and jerk of the host vehicle under driving scenario 1.

Variables	Maximum	Minimum	Average
Acceleration (m/s^2)	0.9702	−0.8606	−0.0331
Jerk (m/s^3)	0.4651	−0.6451	−0.0000163

5.2. Cut In Beyond Safe Distance

In driving scenario 2, there was a potential cut-in vehicle in a nearby lane, since the relative distance between the host vehicle and the nearby vehicle met the safety distance model of the cut-in vehicle, the vehicle did not need to brake in advance. After the cut-in vehicle entered the lane where the vehicle was located, the host vehicle took the cut-in vehicle as the following target. Corresponding speed control was adopted to follow the cut-in vehicle.

In Figure 22, the host vehicle followed the preceding vehicle and maintained a relative distance of 43 m. The nearby vehicle started to cut in at 26 s at a speed of 15 m/s, as shown in Figure 22a. Since the cut-in vehicle safety distance model was satisfied at that time, the host vehicle did not implement braking measures before the cut-in. After the cut-in maneuver was detected by the host vehicle at 26 s, the host vehicle switched to following the cut-in vehicle; thus, the relative distance suddenly changed to 30 m, and the vehicle speed was gradually reduced to 15 m/s, which was the same as the velocity of the preceding vehicle. In Figure 22c,d and Table 2, the maximum deceleration was 0.7310 m/s^2 and the minimum jerk was −0.5383 m/s^3, so the driving safety and comfort were guaranteed during deceleration in this scenario.

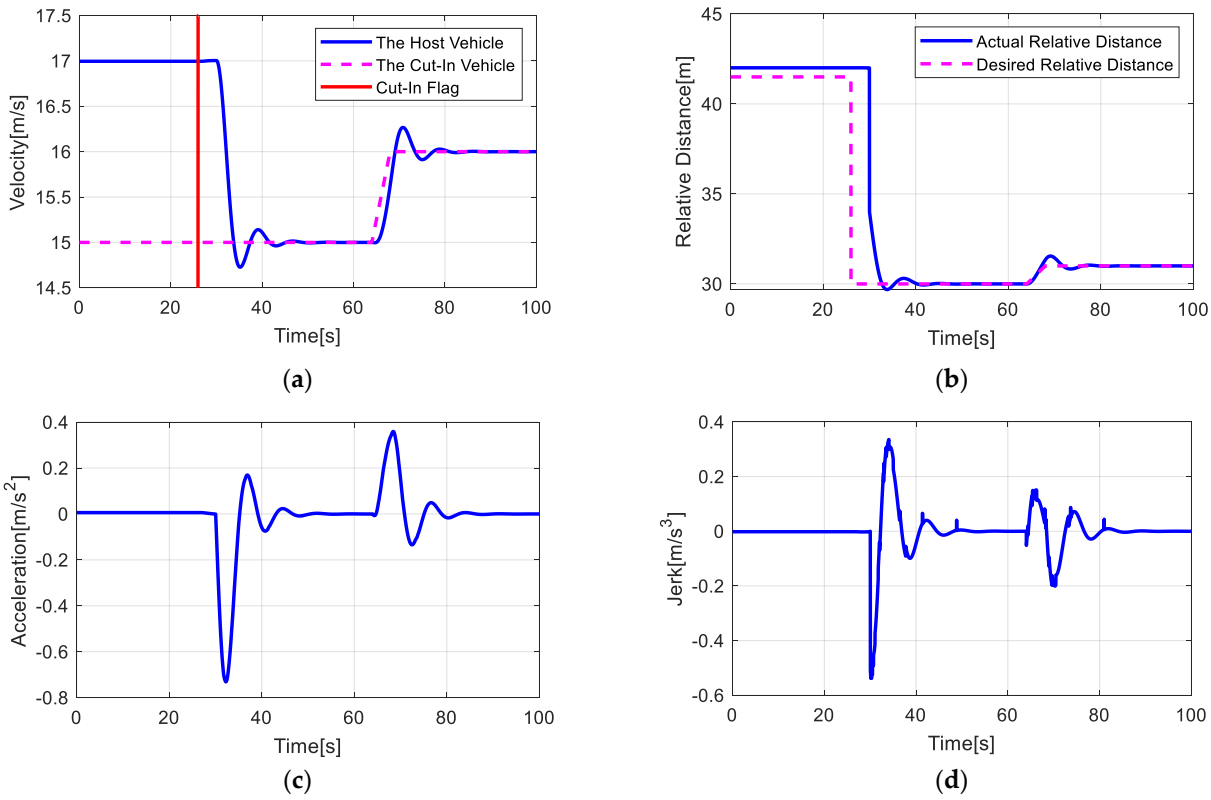

Figure 22. The simulation results of cut-in beyond a safe distance scenario: (**a**) the velocity of the host vehicle and the cut-in vehicle; (**b**) the relative distance between vehicles; (**c**) the acceleration of the host vehicle; and (**d**) the jerk of the host vehicle.

Table 2. Statistics of the acceleration and jerk of the host vehicle under driving scenario 2.

Variables	Maximum	Minimum	Average
Acceleration (m/s^2)	0.3592	−0.7310	−0.0083
Jerk (m/s^3)	0.3357	−0.5383	−0.00052

5.3. Cut-In within Safe Distance

In driving scenario 3, there was a potential cut-in vehicle in a nearby lane and the relative distance between the host vehicle and the nearby vehicle did not satisfy the cut-in

vehicle safety distance model. Therefore, the vehicle needed to brake before the nearby vehicle cut in to provide a sufficient relative distance for the cut-in maneuver of the nearby vehicle, which avoided emergency braking of the vehicle, nervousness of the driver, reduction of driving comfort, and the risk of collision.

In Figure 23, the cut-in vehicle performed a cut-in maneuver at a speed of 15 m/s at 26 s. Because the initial relative distance was 27 m, which did not satisfy the safety distance model of the cut-in vehicle, as shown in Figure 23a, before the nearby vehicle cut in at 26 s, with the cut-in possibility of the nearby vehicle increasing, the host vehicle recognized the cut-in intention of the nearby vehicle and started to slow down at 20 s. The host vehicle speed only decreased to 13.5 m/s which was higher than the minimum speed of the host vehicle without the cut-in recognition. As shown in Figure 23b, due to the early braking of the host vehicle, the relative distance between the host vehicle and the preceding vehicle increased to 43 m, which satisfied the safety distance model of the cut-in vehicle when the nearby vehicle performed the cut-in behavior at 26 s. At the same time, the following target of the host vehicle was smoothly switched to the cut-in vehicle; thus, the relative distance suddenly changed to 27 m, which is close to a stable relative following distance. After several seconds of adjustment, the host vehicle followed the cut-in vehicle steadily. Thus, the cut-in behavior of the nearby vehicle had little effect on the host car. In Figure 23, compared with the host vehicle without cut-in recognition, the vehicle with cut-in recognition obtained a smaller deceleration due to early braking, and prevented the rapid decrease in the relative distance caused by the sudden cut-in behavior of the nearby vehicle, which avoided emergency braking of the vehicle and driver nervousness. As shown in Figure 23c,d, the acceleration and jerk of the host vehicle with cut-in recognition had a smaller variation range, the maximum and minimum of acceleration and jerk of the host vehicle with cut-in recognition is smaller than that of the host vehicle without cut-in recognition, as shown in Tables 3 and 4, so driving safety and comfort were guaranteed.

Table 3. Statistics of the acceleration and jerk of the host vehicle with cut-in recognition under driving scenario 3.

Variables	Maximum	Minimum	Average
Acceleration (m/s^2)	0.5963	−1.2806	−0.0104
Jerk (m/s^3)	0.9513	−0.7068	−0.00042

Table 4. Statistics of the acceleration and jerk of the host vehicle without cut-in recognition under driving scenario 3.

Variables	Maximum	Minimum	Average
Acceleration (m/s^2)	1.5597	−1.4305	−0.0102
Jerk (m/s^3)	1.0805	−0.7795	−0.00052

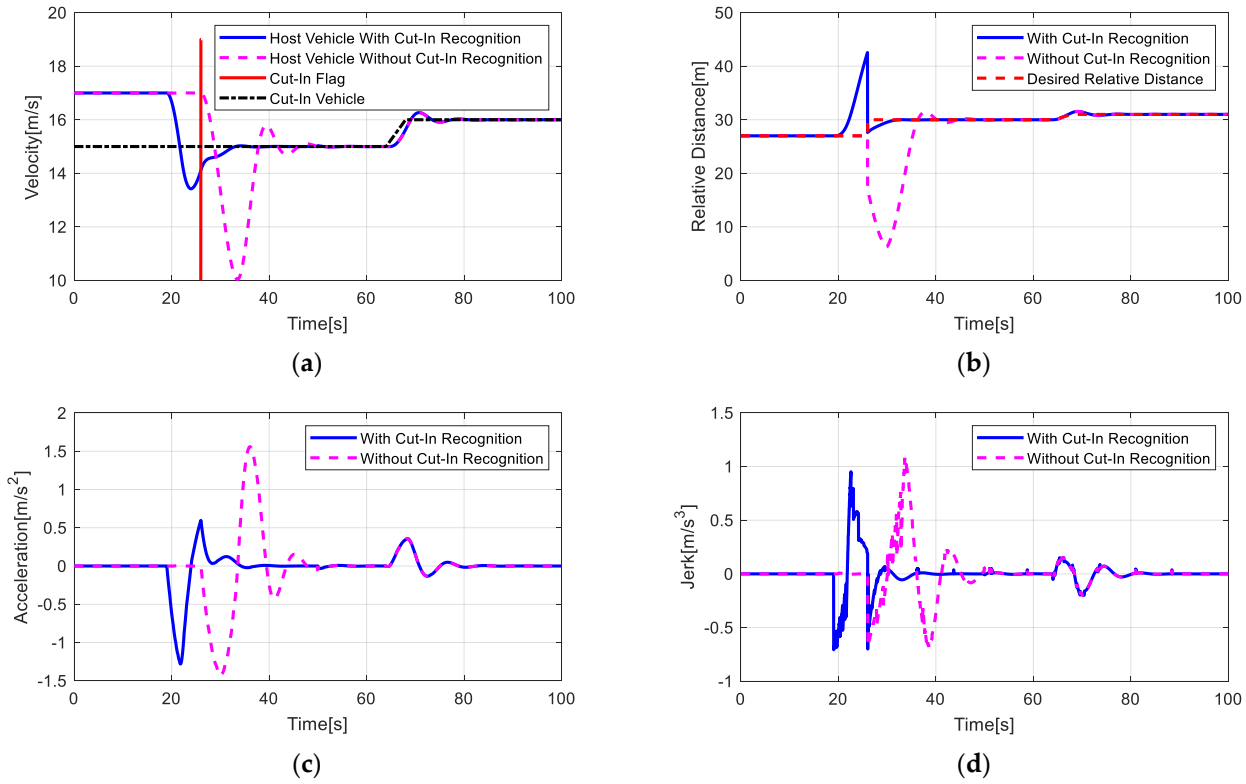

Figure 23. The simulation results of cut-in within a safe distance scenario: (**a**) the velocity comparison of the host vehicle with cut-in recognition and the host vehicle without cut-in recognition; (**b**) the following distance comparison of the host vehicle with cut-in recognition and the host vehicle without cut-in recognition; (**c**) the acceleration comparison; and (**d**) the jerk comparison.

5.4. The Complexity of MPC–ACC

Considering the computational complexity of MPC, the complexity of the model needs to be discussed. The simulation test of driving scenario 3 was selected to calculate the complexity of the model.

As shown in Table 5, the simulation termination time was set to 100 s, but the model actually took 48 s to run. A total of 8535 MB of RAM was used during the running of the model, so the developed method is suitable for real-time applications.

Table 5. Statistics of the complexity of the model under driving scenario 3.

Variables	Value
Simulation termination time (s)	100
Actual simulation time (s)	48.62
RAM (MB)	8535

6. Conclusions

This paper proposes an ACC for cut-in scenarios based on the model predictive control algorithm. Through the screening of potential cut-in vehicles, the cut-in possibility of a nearby vehicle was quantified, which was used as a reference for the MPC to coordinate the control of the host vehicle. The FSM was used to divide the cut-in scenarios, and the established safe distance model of the cut-in vehicle was used to manage the state transition of the FSM. Using the cut-in vehicle under different scenarios, different reference trajectories were generated by the FSM for the MPC–ACC control. Under the constraints of driving safety and comfort, the MPC was used to solve the optimal control trajectory of the host vehicle with different reference trajectories, which determined that the host

vehicle can follow the cut-in vehicle safely and comfortably. The proposed control strategy for cut-in scenarios was validated using the Simulink–Carsim ® simulation platform. For future research, more cut-in scenarios will be investigated and validated; furthermore, experimental verifications will be conducted.

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